XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	COM205		
Course Name:	Microelectronics Devices and Circuits		
Question Paper Setter:	Shams Immad		
Academic Session:	2021/09	Question Paper:	A ☐ B ☑
Total No. of Pages:	5	Time Allocated:	2 hours
Additional Materials:	-		
Apparatus Allowed:	Calculator		

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 5 Questions:

Question 1:	8	questions	Question 2:	7	questions
Question 3:	4	questions	Question 4:	1	question
Question 5:	1	question			

   Answer all questions.

2. Please follow the requirement of each section and type in all the corresponding answers on the downloadable answer book provided.

3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (20 marks) **[CLO1-PLO1:C2]**

Given the BJT transistor characteristics of Fig. 1:

(a) Draw a load line on the characteristics determined by voltage, V_{CC} = 21 V and collector resistance, R_C = 3 kΩ for a fixed-bias configuration.

[4 marks]

(b) Choose an operating point midway between cutoff and saturation. Determine the value of R_B to establish the resulting operating point.

[4 marks]

(c) What are the resulting values of I_{CQ} and V_{CEQ}?

[2 marks]

(d) What is the value of β at the operating point?

[2 marks]

(e) What is the value of α defined by the operating point?

[2 marks]

(f) What is the dc power dissipated by the device at the operating point?

[2 marks]

(g) What is the power supplied by V_{CC}?

[2 marks]

(h) Determine the power dissipated by the resistive elements by taking the difference between the results of parts (f) and (g)?

[2 marks]

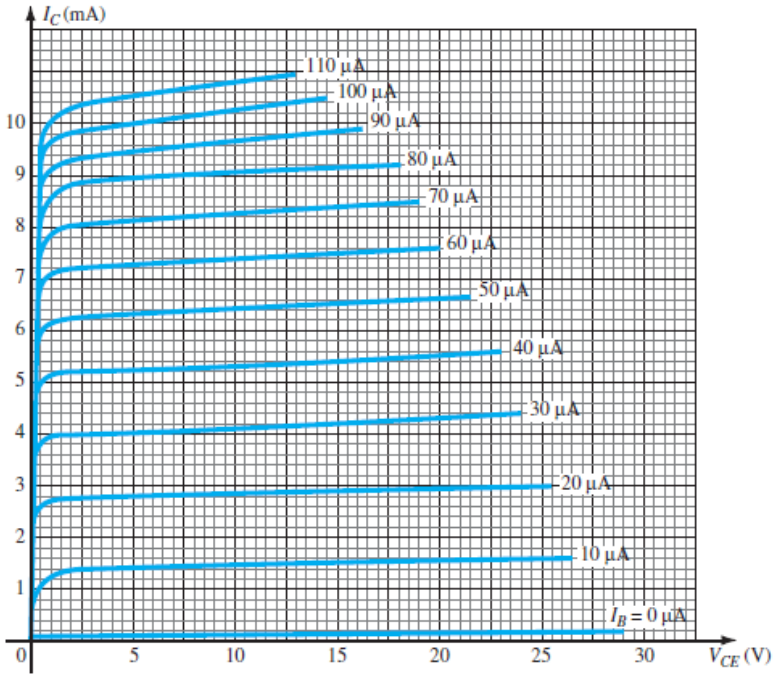


Fig. 1: Transistor characteristics curve.

Question 2 (25 marks) **[CLO2-PLO2:C4]**

For the network of Fig. 2: -

(a) Determine V_{GSQ} and I_{DQ}.

[7 marks]

(b) Find g_{mO} and g_m.

[4 marks]

(c) Calculate the midband gain of $A_v = \frac{V_o}{V_i}$.

[2 marks]

(d) Determine Z_i.

[2 marks]

(e) Calculate $A_{vs} = \frac{V_o}{V_s}$.

[2 marks]

(f) Determine f_{LG}, f_{LC} and f_{LS}.

[6 marks]

(g) Determine the low-cutoff frequency.

[2 marks]

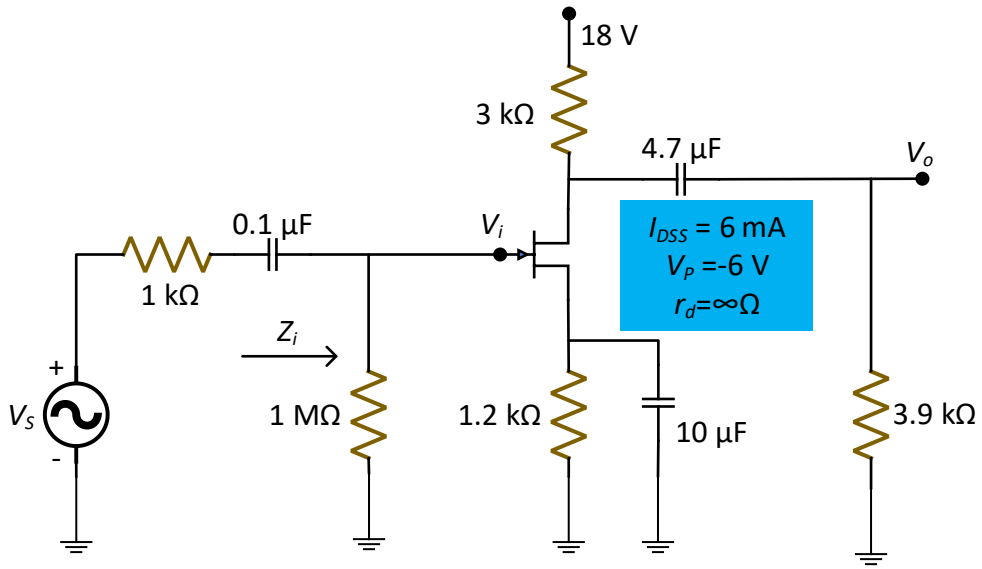


Fig. 2: Self biased FET amplifier.

Question 3 (20 marks) **[CLO2-PLO2:C4]**

(a) Calculate the output voltage developed by the circuit of Fig. 3 for R_f = 330 kΩ.

[6 marks]

(b) Calculate the output voltage of the circuit in Fig. 3 for R_f = 68 kΩ.

[6 marks]

(c) Explain the effect of change in R_f between (a) and (b).

[2 marks]

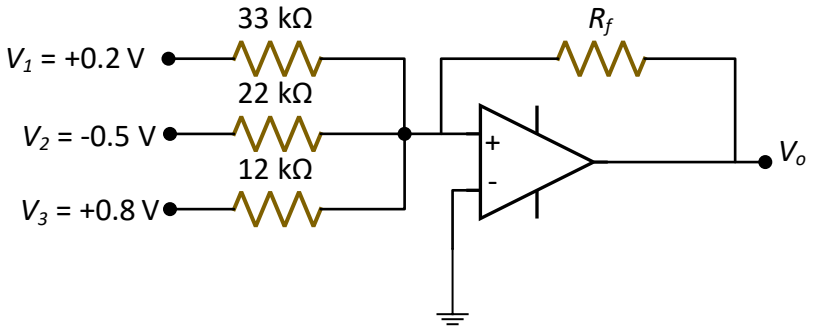


Fig. 3: Op-amp circuit.

(d) Calculate the output voltage V_2 and V_3 in the circuit of Fig. 4.

[6 marks]

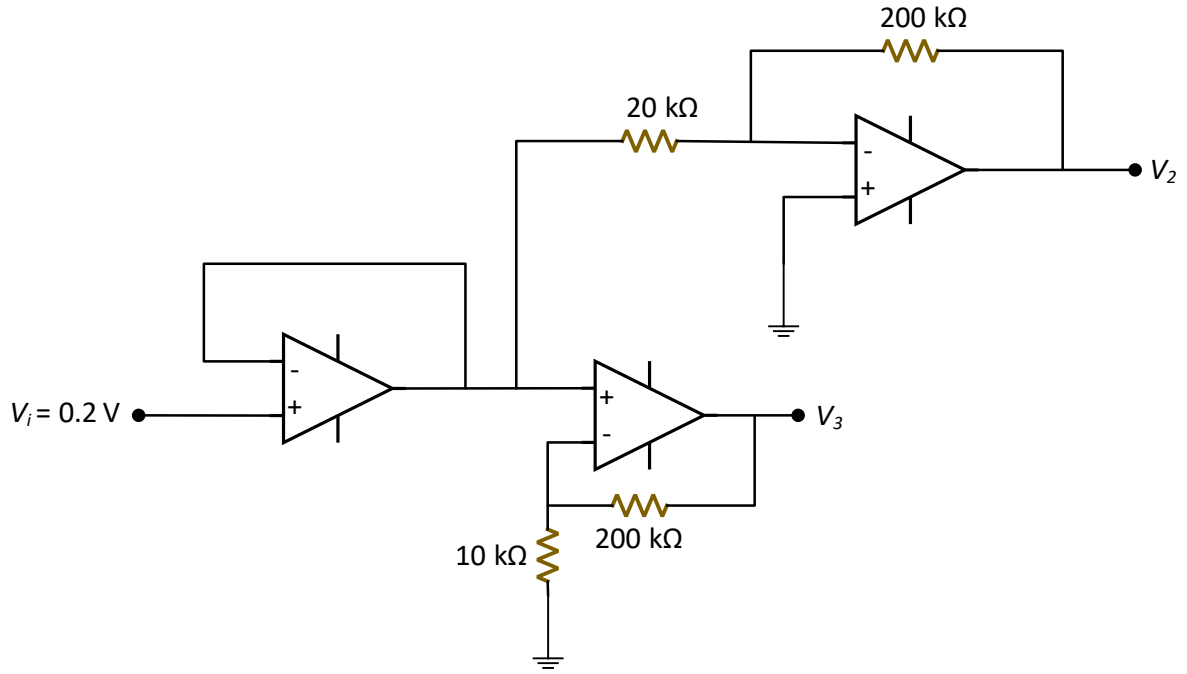


Fig. 4: Combination of Op-amp circuits.

Question 4 (15 marks) **[CLO3-PLO3:C6]**

(a) Design a voltage-divider bias network using a depletion-type MOSFET with I_{DSS}=10 mA and V_P = -4 V to have a Q-point at I_{DQ}=2.5 mA using a supply of 24 V. In addition, set V_G=4V and use R_D=2.5R_S with R_1=22 MΩ. Use standard values.

Question 5 (20 marks) **[CLO3-PLO3:C6]**

(a) For the network of Fig. 5, design R_E and R_B if A_v=-10 and r_e=3.8 Ω. Assume that $Z_b=\beta R_E$.

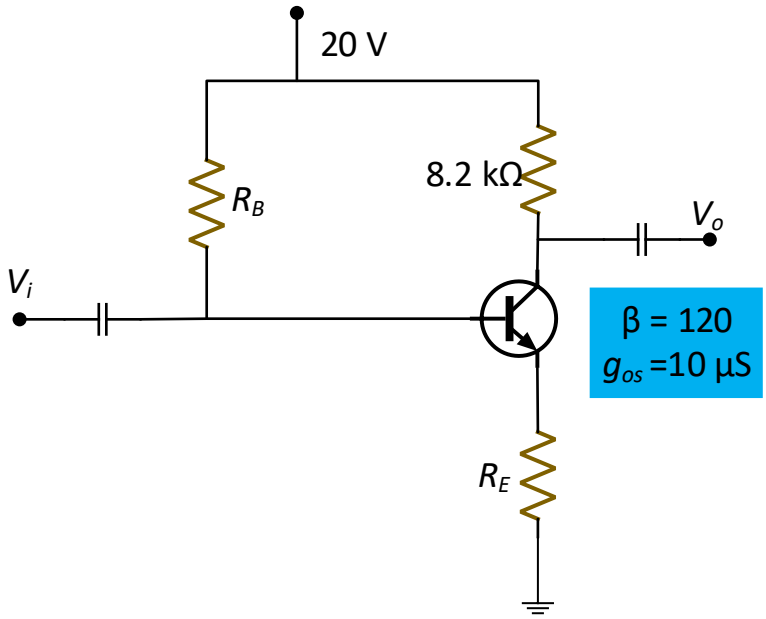


Fig. 5. Self-bias configuration.

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